ProveIt Database Schema

This document describes the current MongoDB schema used by the ProveIt platform. It was generated from the Mongoose models in backend/src/models.

Database

- Database type: MongoDB
- ODM: Mongoose
- Main connection: backend/src/lib/db.js
- Connection environment variable: DB_URL

Collections Overview

Collection	Mongoose Model	Purpose
users	User	Stores application users, Clerk identity mapping, score, level, and role.
subscriptions	Subscription	Stores plan, billing period, and AI interview limits per user.
sessions	Session	Stores live coding sessions, host/participant state, difficulty, selected problem, and Stream call id.
invitations	Invitation	Stores scheduled session invitations and acceptance/decline state.
notifications	Notification	Stores in-app notifications for invitations and session events.
availabilities	Availability	Stores weekly availability slots for scheduling.
codesubmissions	CodeSubmission	Stores Judge0-backed problem submission attempts and score deltas.
candidateprofiles	CandidateProfile	Stores AI CV analysis and adaptive interview progress per user.
aiinterviewsessions	AIInterviewSession	Stores AI interview attempts, questions, answers, scores, and final reports.
interviewquestions	InterviewQuestions	Stores seeded interview question bank records for RAG/question selection.
blogposts	BlogPost	Stores localized public/admin blog posts.

Entity Relationship Summa

```
User 1--1 Subscription
User 1--1 Availability
User 1--1 CandidateProfile
User 1--N CodeSubmission
User 1--N AIInterviewSession
User 1--N Session as host
User 1--N Session as participant/invitedP
User 1--N Invitation as sender
User 1--N Invitation as recipient
User 1--N Notification as recipient/sende
Invitation 0--1 Session
Session 0--1 Invitation
AIInterviewSession N--N InterviewQuestion
```

User

Model file: backend/src/models/User.js

Field	Type	Required	Default	Notes
name	String	Yes	none	Display name.
email	String	Yes	none	Unique.
profileImage	String	No	""	Avatar URL.
clerkId	String	Yes	none	Unique Clerk user id.
role	String enum	No	user	user, admin.
score	Number	No	0	Minimum 0.
level	String enum	No	Entry-Level	Entry-Level, Junior, Intermediate, Advanced.
scoreStats.easySolved	Number	No	0	Minimum 0.
scoreStats.mediumSolved	Number	No	0	Minimum 0.
scoreStats.hardSolved	Number	No	0	Minimum 0.
scoreStats.totalSolved	Number	No	0	Minimum 0.
scoreStats.successRate	Number	No	0	0 to 100.
scoreStats.lastActivityBonusDate	String	No	""	Daily bonus date key.
scoreStats.lastUpdated	Date	No	Date.now	Stats update time.
createdAt	Date	Auto	auto	Mongoose timestamp.
updatedAt	Date	Auto	auto	Mongoose timestamp.

Indexes and constraints:

- Unique: email
- Unique: clerkId

Subscription

Model file: backend/src/models/

Field	Type	Required	Default	Notes
userId	ObjectId ref User	Yes	none	Unique subscription owner.
plan	String enum	No	free	free, pro, premium.
status	String enum	No	active	active, cancelled, expired.
currentPeriodStart	Date	No	none	Billing period start.
currentPeriodEnd	Date	No	none	Billing period end.
aiInterviewsUsedThisMonth	Number	No	0	Monthly usage counter.

aiInterviewsLimit	Number	No	3	-1 means unlimited in plan logic.
cancelledAt	Date	No	null	Cancellation time.
createdAt	Date	Auto	auto	Mongoose timestamp.
updatedAt	Date	Auto	auto	Mongoose timestamp.

Indexes and constraints:

- Unique: userId

Session

Model file: backend/src/models/

Field	Type	Required	Default	Notes
problem	String	No	""	Current selected problem title.
difficulty	String enum	Yes	none	easy, medium, hard.
host	ObjectId ref User	Yes	none	Session creator/host.
participant	ObjectId ref User	No	null	User who joined or accepted invitation.
invitedParticipant	ObjectId ref User	No	null	Reserved invited user for private sessions.
invitation	ObjectId ref Invitation	No	null	Source invitation.
selectedProblemId	String	No	""	Current selected problem id.
startedNotificationSentAt	Date	No	null	Host-started notification marker.
participantStartedNotificationSentAt	Date	No	null	Participant-started notification marker.
status	String enum	No	active	active, completed.
callId	String	Yes	none	Unique Stream video/ chat id.
createdAt	Date	Auto	auto	Mongoose timestamp.
updatedAt	Date	Auto	auto	Mongoose timestamp.

Indexes and constraints:

- Unique: callId
- Index: { status: 1, createdAt: -1
- Index: { host: 1 }
- Index: { participant: 1 }
- Index: { invitedParticipant: 1 }

Invitation

Model file: backend/src/models/

Field	Type	Required	Default	Notes
senderId	ObjectId ref User	Yes	none	Inviter.
recipientId	ObjectId ref User	Yes	none	Invitee.
scheduledDate	String	Yes	none	Format YYYY-MM-DD.
scheduledTime	String	Yes	none	Format HH:MM.
duration	Number	No	60	Minutes.
problem	String	No	""	Legacy/optional problem title.
difficulty	String enum	Yes	none	easy, medium, hard.
message	String	No	""	Optional sender note.
status	String enum	No	pending	pending, accepted, declined.
sessionId	ObjectId ref Session	No	null	Created after acceptance.
createdAt	Date	Auto	auto	Mongoose timestamp.
updatedAt	Date	Auto	auto	Mongoose timestamp.

Indexes:

- { recipientId: 1, status: 1 }
- { senderId: 1 }

Notification

Model file: backend/src/models/

Field	Type	Required	Default	Notes
recipientId	ObjectId ref User	Yes	none	Notification owner.
senderId	ObjectId ref User	Yes	none	Notification actor.
type	String enum	Yes	none	session-invitation, invitation-accepted, invitation-declined, session-reminder, session-started.
status	String enum	No	unread	unread, read.
relatedInvitationId	ObjectId ref Invitation	No	null	Related invitation.
relatedSessionId	ObjectId ref Session	No	null	Related session.
message	String	No	""	Display message.
createdAt	Date	Auto	auto	Mongoose timestamp.
updatedAt	Date	Auto	auto	Mongoose timestamp.

Indexes:

- { recipientId: 1, status: 1 }

Availability

Model file: backend/src/models/Availability.js

Field	Type	Required	Default	Notes
userId	ObjectId ref User	Yes	none	Unique owner.
weeklySchedule.monday	Slot[]	No	[]	Time ranges.
weeklySchedule.tuesday	Slot[]	No	[]	Time ranges.
weeklySchedule.wednesday	Slot[]	No	[]	Time ranges.
weeklySchedule.thursday	Slot[]	No	[]	Time ranges.
weeklySchedule.friday	Slot[]	No	[]	Time ranges.
weeklySchedule.saturday	Slot[]	No	[]	Time ranges.
weeklySchedule.sunday	Slot[]	No	[]	Time ranges.
timezone	String	No	Africa/Cairo	Scheduling timezone.
updatedAt	Date	No	Date.now	Explicit update marker plus timestamps.
createdAt	Date	Auto	auto	Mongoose timestamp.

Slot subdocument:

Field	Type	Required	Notes
startTime	String	Yes	Format HH:MM.
endTime	String	Yes	Format HH:MM.

Indexes and constraints:

- Unique: userId

CodeSubmission

Model file: backend/src/models/CodeS

Field	Type	Required	Default	Notes
userId	ObjectId ref User	Yes	none	Submitter.
problemId	String	Yes	none	Practice problem id.
problemTitle	String	Yes	none	Practice problem title.
difficulty	String enum	Yes	none	easy, medium, hard.
language	String	No	""	javascript, python, or java from frontend runner.
accepted	Boolean	Yes	none	Whether output matched expected output.
firstTry	Boolean	No	false	Accepted with no previous attempts.

solvedPointsAwarded	Number	No	0	Easy 10, medium 20, hard 35.
firstTryBonusAwarded	Number	No	0	5 when applicable.
dailyBonusAwarded	Number	No	0	3 when applicable.
wrongPenaltyApplied	Number	No	0	-2 once per problem for wrong attempt.
scoreDelta	Number	No	0	Total score impact of this submission.
createdAt	Date	Auto	auto	Mongoose timestamp.
updatedAt	Date	Auto	auto	Mongoose timestamp.

Indexes:

- { userId: 1, problemId: 1, creat
- { userId: 1, accepted: 1 }

CandidateProfile

Model file: backend/src/models/

Field	Type	Required	Default	Notes
userId	ObjectId ref User	Yes	none	Unique owner.
primaryField	String	No	fullstack	CV-inferred field.
yearsOfExperience	Number	No	0	CV-inferred experience.
keySkills	String[]	No	[]	CV skills.
strongestAreas	String[]	No	[]	CV strengths.
notableTechnologies	String[]	No	[]	CV technologies.
cvSummary	String	No	""	Summary fed to AI interview.
preferredDifficulty	String enum	No	medium	easy, medium, hard.
questionsAskedIds	ObjectId[] ref InterviewQuestions	No	[]	Prevents repeats.
totalInterviewsCompleted	Number	No	0	Progress metric.
overallAverageScore	Number	No	0	Progress metric.
topicWeaknesses	Object	No	{}	Adaptive topic tracking.
topicStrengths	Object	No	{}	Adaptive topic tracking.
createdAt	Date	Auto	auto	Mongoose timestamp.
updatedAt	Date	Auto	auto	Mongoose timestamp.

Indexes and constraints:

- Unique: userId

AIInterviewSessio

Model file: backend/src/models/AIInterviewSession.js

Field	Type	Required	Default	Notes
userId	ObjectId ref User	Yes	none	Candidate.
cvAnalysis	Object	No	{}	Snapshot of candidate profile.
language	String enum	No	en	en, ar.
interviewerName	String	No	Zara	AI interviewer.
currentStage	String	No	intro	Interview stage.
targetRole	String	No	""	Optional target role.
jobDescription	String	No	""	Optional job description.
questions	Question[]	No	[]	Asked questions and answers.
status	String enum	No	in-progress	in-progress, completed.
overallScore	Number	No	0	Final score.
overallFeedback	String	No	""	Final feedback.
strengths	String[]	No	[]	Final strengths.
weaknesses	String[]	No	[]	Final weaknesses.
recommendation	String	No	""	Final recommendation.
reportSummary	String	No	""	Final report summary.
stageHistory	Object[]	No	[]	Stage transitions.
createdAt	Date	Auto	auto	Mongoose timestamp.
updatedAt	Date	Auto	auto	Mongoose timestamp.

Question subdocument:

Field	Type	Required	Default	Notes
questionId	ObjectId ref InterviewQuestions	No	none	Optional question bank link.
questionText	String	Yes	none	Spoken question.
topic	String	No	""	Topic tag.
difficulty	String	No	""	easy, medium, hard.
stage	String	No	technical	Interview stage.
userAnswer	String	No	""	Transcribed answer.
aiFeedback	String	No	""	Per-answer feedback.
score	Number	No	0	0 to 10.
timeSpent	Number	No	0	Seconds.
followUpAsked	Boolean	No	false	Follow-up flag.

followUpAnswer	String	No	""	Follow-up answer.

Indexes:

- { userId: 1, createdAt: -1 }

InterviewQuestion

Model file: backend/src/models/

Field	Type	Required	Default	Notes
questionText	String	Yes	none	Question prompt.
category	String	Yes	none	Topic category.
fieldRelevance	String[]	Yes	none	Relevant candidate fields.
difficulty	String enum	Yes	none	easy, medium, hard.
followUpQuestions	String[]	No	[]	Suggested follow-ups.
expectedKeywords	String[]	No	[]	Evaluation hints.
createdAt	Date	Auto	auto	Mongoose timestamp.
updatedAt	Date	Auto	auto	Mongoose timestamp.

Indexes:

- fieldRelevance indexed throug
- { fieldRelevance: 1, difficulty: 1

BlogPost

Model file: backend/src/models/

Localized text subdocument:

Field	Type	Default
en	String	""
ar	String	""

Blog section subdocument:

Field	Type	Notes
heading	LocalizedText	Section heading.
body	LocalizedText	Section body.

Blog post document:

Field	Type	Required	Default	Notes
title	LocalizedText	No	localized empty strings	Title in EN/AR.
slug	String	Yes	none	Unique, lowercase.
excerpt	LocalizedText	No	localized empty strings	Listing excerpt.

category	LocalizedText	No	localized empty strings	Post category.
readTime	LocalizedText	No	localized empty strings	Read-time label.
author	LocalizedText	No	localized empty strings	Author label.
content	BlogSection[]	No	[]	Localized article sections.
published	Boolean	No	false	Public visibility.
publishedAt	Date	No	null	Publication date.
createdAt	Date	Auto	auto	Mongoose timestamp.
updatedAt	Date	Auto	auto	Mongoose timestamp.

Indexes and constraints:

- Unique: slug
- Index: { published: 1, published
- Index: { slug: 1 }, unique

Scoring Rules

Source: backend/src/services/sc

Rule	Value
Easy solved points	10
Medium solved points	20
Hard solved points	35
First try bonus	5
Daily activity bonus	3
Wrong submission penalty	-2

User level thresholds:

Score Range	Level
0 to 99	Entry-Level
100 to 199	Junior
200 to 299	Intermediate
300+	Advanced

Notes

- Practice problems are currently frontend data in frontend/src/data
CodeSubmission.
- Session video/chat state is created in Stream using callId; Mongo
- Authentication identity is owned by Clerk; MongoDB stores the ap